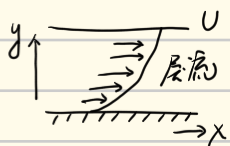


流体的性质: 黏性 (Viscosity) $\mu, \text{Pa}\cdot\text{s}$ (运动状态表现) $\rightarrow$ 阻力特征

牛顿流体: $F = \mu A \frac{U}{h}$ $\tau = \frac{F}{A} = \mu \frac{U}{h}$
 满足牛顿内摩擦定律 $\downarrow$ 速度梯度 $\downarrow$ 应力 $\downarrow$ 线性关系



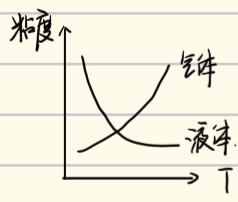
雷诺数 Re 判断层流和湍流

$Re = \frac{\rho U L}{\mu}$ 无量纲数, 代表流动性

$Re \leq 2300$ 阻力 $\propto U$
 $Re > 2300$ 阻力 $\propto U^{1.75} \sim U^2$

影响粘性的因素

气体粘性取决于分子无规则运动 $\rightarrow$ 随温度增加增加
 液体粘性取决于分子间引力 $\rightarrow$ 随温度增加减小



牛顿内摩擦规律

$\tau = \mu \frac{du}{dy}$ $\frac{du}{dy} \neq 0 \rightarrow$ 粘性将以内摩擦形式表现出来 $\rightarrow$ 改变流体流动

动力粘度 μ , 单位 $\text{Pa}\cdot\text{s}$

运动粘度 ν , $\nu = \frac{\mu}{\rho}$, 单位 m^2/s

$Re = \frac{UL}{\nu}$

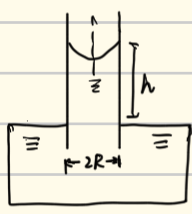
表面张力



表面张力是作用在界面上的力

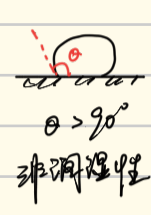
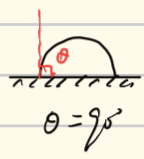
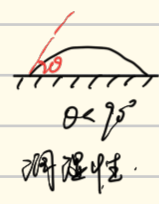
$F = \gamma L$

表面张力系数 γ



$h = \frac{2\gamma \cos \theta}{\rho g R}$

接触角



基本的流动分析方法

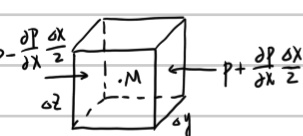
- 1) 控制体积法/积分分析法
- 2) 无量纲系统法/微分分析法
- 3) 实验方法/量纲分析法

流体静力学

Contents of this chapter

- 1) 大气和液体中的压力分布 (pressure distribution)
- 2) 压力计的设计 (manometer)
- 3) 施加于浸没平板和曲面上的力 (pressure on merging surface)
- 4) 浸没物体上的浮力 (buoyancy) } result in Archimedes' principles
- 5) 浮体的性质

2.1 流体平衡微分方程



$$f_x = \frac{1}{\rho} \frac{\partial p}{\partial x} \quad f_y = \frac{1}{\rho} \frac{\partial p}{\partial y} \quad f_z = \frac{1}{\rho} \frac{\partial p}{\partial z} \quad \vec{f} = \frac{1}{\rho} \nabla p \quad \nabla = \vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z}$$

$$dp = \frac{\partial p}{\partial x} dx + \frac{\partial p}{\partial y} dy + \frac{\partial p}{\partial z} dz = \rho (f_x dx + f_y dy + f_z dz) \quad \text{压强差公式}$$

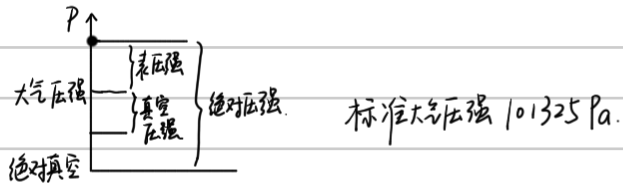
2.2 重力作用下静止流体中的压强分布

$$f_x = 0 \quad f_y = 0 \quad f_z = -g$$

$$dp = -\rho g dz \Rightarrow p = -\rho g z + C \quad \text{或} \quad z + \frac{p}{\rho g} = C$$

$$z_1 + \frac{p_1}{\rho g} = z_2 + \frac{p_2}{\rho g}$$

$$p = p_0 + \rho g (z_0 - z) = p_0 + \rho g h \quad \text{帕斯卡原理}$$



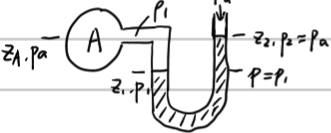
液柱测压计

U形管测压计

$$p_a = -\nabla p + \rho g + \mu \nabla^2 v$$

驱动力 表面力 体积力 粘性力(切向)
(法向)

只在重力作用下, 则 $p_a = 0, \mu \nabla^2 v = 0 \Rightarrow \nabla p = \rho g$



3. 在重力作用下可压缩流体的压强分布

国际标准大气的温度分布

$$\begin{cases} T = 288 - 0.0065z \text{ K}, & z \leq 11000 \text{ m} & \text{对流层} \\ T = 216.5 \text{ K}, & z > 11000 \text{ m} & \text{同温层} \end{cases}$$

完全气体状态方程 $p = \rho R T$ 对于空气, $R = 287 \text{ J/(kg} \cdot \text{K)}$

1) 对流层压强分布

$$p = \frac{P}{RT} = \frac{P}{R(288 - 0.0065z)} \Rightarrow \frac{dp}{p} = \frac{-g dz}{R(288 - 0.0065z)} \Rightarrow \frac{p}{p_a} = \left(1 - \frac{0.0065z}{288}\right)^{\frac{g}{0.0065R}} \Rightarrow p = p_a \left(1 - \frac{z}{44308}\right)^{5.256} \quad (z \leq 11000 \text{ m})$$

当 $z = 11000 \text{ m}$ 时, $p_i = 0.223 p_a$

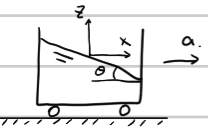
2) 同温层中压强分布

$$p = \frac{P}{RT} = \frac{P}{216.5 R} \Rightarrow \frac{p}{p_i} = \exp\left[\frac{g(z - 11000)}{216.5 R}\right] \Rightarrow p = 0.223 p_a \exp\left(-\frac{z - 11000}{6336}\right) \quad (z > 11000 \text{ m})$$

2.3 相对静止液体中的压强分布

1. 水平步加速运动液体在重力作用下的压强分布

$$\nabla p = \rho(\vec{g} - \vec{a})$$



$$f_x = -a, f_y = 0, f_z = -g$$

$$dp = \rho(f_x dx + f_y dy + f_z dz) = \rho(-a dx - g dz)$$

$$\Rightarrow p = \rho(-ax - gz) + C$$

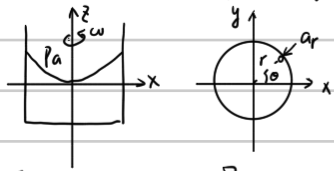
在 $x=0, z=0$ 处, $p=p_a$, 则有 $p=p_a - \rho(ax+gz)$

令 p 为常数, 则有 $ax+gz=C \Rightarrow$ 等压面方程. 等压面与水平面夹角 $\theta = \arctan \frac{a}{g}$

自由面也是等压面. 自由面方程 $z_0 = -\frac{a}{g}x$

相对压强 $p-p_a = \rho g(z_0-z) = \rho gh$

2. 角速度旋转液体在重力作用下的压强分布



$$f_x = \omega^2 x, f_y = \omega^2 y, f_z = -g$$

$$dp = \rho(\omega^2 x dx + \omega^2 y dy - g dz)$$

$$\Rightarrow p = \rho\left(\frac{\omega^2}{2}x^2 + \frac{\omega^2}{2}y^2 - gz\right) + C = \rho\left(\frac{\omega^2}{2}r^2 - gz\right) + C$$

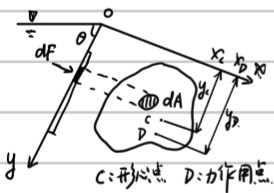
原点取在自由面中心最低点上. $p = p_a + \rho g\left(\frac{\omega^2}{2g}r^2 - z\right)$

等压面方程: $\frac{\omega^2}{2}x^2 + \frac{\omega^2}{2}y^2 - gz = \frac{\omega^2}{2}r^2 - gz = C$. 自由面方程: $z_0 = \frac{\omega^2}{2g}r^2$. $p-p_a = \rho g(z_0-z) = \rho gh$.

当液面上有盖子时, 压强变化规律仍适用.

2.4 静止液体作用在物体壁面上的总压力

1. 平面壁面



$$dF = pgh dA = \rho g y \sin \theta dA$$

$\rightarrow$ 面积矩 (A 对 x 轴的)

$$F = \int_A dF = \rho g \sin \theta \int_A y dA \quad \int_A y dA = y_c A$$

$$F = \rho g y_c \sin \theta \cdot A = \rho g h_c A \quad h_c \text{ — 形心点水深.}$$

$$dF = \rho g y \sin \theta dA \quad \int_A x dF = \rho g \sin \theta \int_A xy dA = \rho g \sin \theta J_{xy}$$

$$F = \rho g y_c \sin \theta \cdot A \quad \int_A y dF = \rho g \sin \theta \int_A y^2 dA = \rho g \sin \theta J_x$$

$$J_{xy} = J_{cy} + x_c y_c A \quad J_x = J_{cx} + y_c^2 A \quad \text{平行移轴定理}$$

$$\text{主惯性矩} \quad x_D - x_c = \frac{J_{cx}}{y_c A}$$

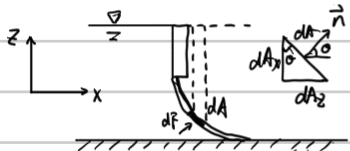
$$\text{主惯性矩} \quad y_D - y_c = \frac{J_{cy}}{x_c A}$$

压力中心位置总在形心点位置之下.

常见图形的 A , y_c 及 J_{xc} 值			
几何图形名称	面积 A	形心坐标 y_c	对通过形心轴的惯性矩 J_{xc}
矩形 	bh	$\frac{1}{2}h$	$\frac{1}{12}bh^3$
三角形 	$\frac{1}{2}bh$	$\frac{2}{3}h$	$\frac{1}{36}bh^3$
圆 	πr^2	r	$\frac{1}{4}\pi r^4$

2. 曲面壁面

(1) 水平分力 F_x



$$F_x = \int_A \cos \theta dF = \int_A \rho g h \cos \theta dA$$

$$F_x = \rho g \int_{A_x} h dA_x = \rho g h_{xc} A_x$$

A_x — 曲面 A 在 x 方向的投影面积.

h_{xc} 是面积 A_x 形心点的水深.

(2) 垂直分力 F_z

$$F_z = \int_A dF_z = \int_{A_z} \rho g h dA_z = \rho g \int_{A_z} h dA_z = \rho g V \quad V \text{ — 压力体体积}$$

$$\text{合力 } F = \sqrt{F_x^2 + F_z^2} \quad \tan \alpha = \frac{F_z}{F_x}$$

理想流体动力学基础

3.1 描述流体运动的两种方法

1. 描述流体运动的方法

(1) 拉格朗日法 质点—时间描述法

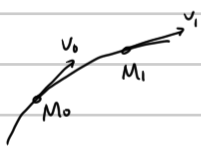
$$\text{空间位置} \begin{cases} x = x(\xi, \eta, \zeta, t) \\ y = y(\xi, \eta, \zeta, t) \\ z = z(\xi, \eta, \zeta, t) \end{cases}$$

速度、加速度求偏导

(2) 欧拉法 空间—时间描述法

$$u = u(x, y, z, t), v = v(x, y, z, t), w = w(x, y, z, t), \rho = \rho(x, y, z, t), p = p(x, y, z, t)$$

2. 质点加速度与质点导数



$$v_0(x, y, z, t)$$

$$v_1(x + \Delta x, y + \Delta y, z + \Delta z, t + \Delta t)$$

$$a = \lim_{\Delta t \rightarrow 0} \frac{v_1 - v_0}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t} = \lim_{\Delta t \rightarrow 0} \left(\frac{\partial v}{\partial t} + \frac{\partial v}{\partial x} \frac{\Delta x}{\Delta t} + \frac{\partial v}{\partial y} \frac{\Delta y}{\Delta t} + \frac{\partial v}{\partial z} \frac{\Delta z}{\Delta t} \right) = \frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z}$$

$$\vec{a} = \vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z} \quad \vec{a} = \frac{\partial \vec{v}}{\partial t} + (\vec{v} \cdot \nabla) \vec{v}$$

$$\begin{cases} a_x = \frac{du}{dt} = \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} \\ a_y = \frac{dv}{dt} = \frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} \\ a_z = \frac{dw}{dt} = \frac{\partial w}{\partial t} + u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} \end{cases}$$

局部加速度 + 对流加速度

质点导数 = 局部导数 + 对流导数

$$\frac{d}{dt} = \frac{\partial}{\partial t} + u \frac{\partial}{\partial x} + v \frac{\partial}{\partial y} + w \frac{\partial}{\partial z}$$

3. 流动的分类

定常流动 $u = u(x, y, z), v = v(x, y, z), w = w(x, y, z)$

$$\frac{du}{dt} = 0, \frac{dv}{dt} = 0, \frac{dw}{dt} = 0 \quad \text{局部加速度为0, 但对流加速度不一定为0. } \frac{d}{dt} \text{ 不一定为0}$$

一元流动 只与一个变量有关

二元流动 (平面流动与轴对称流动)

3.2 迹线、流线、流管

1. 迹线: 流体质点的运动轨迹线

流线: 流场空间中与速度矢量处处相切的曲线

脉线 (染色线): 相继通过流场同一空间点的流体质点所连成的曲线

设 $d\vec{s} = dx\vec{i} + dy\vec{j} + dz\vec{k}$ 为流线上一点的微元弧长

$\vec{v} = u\vec{i} + v\vec{j} + w\vec{k}$ 为流质在 A 点的流速

$$\text{流线方程: } \frac{dx}{u} = \frac{dy}{v} = \frac{dz}{w}$$

流管: 由流线构成的管状曲面

流束: 流管内流体

总流: 截面积有限的流束整体

2. 流量

体积流量 $Q = \int_A v_n dA$

质量流量 $Q_m = \int_A \rho v_n dA$

1. 质量守恒方程 $\rightarrow$ 体积流量 $\times$ 密度

2. 动量守恒方程

3. 能量守恒方程

4. 气体状态方程

流面积上的平均速度: $V = \frac{Q}{A}$

3.3 连续性方程

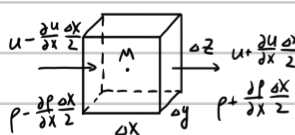
1. 质量系统与控制体

质量系统: 由特定流体质点组成的流体团.

控制体: 流场中人为定义的空间区域.

类似拉格朗日法与欧拉法.

2. 微分形式的连续性方程 (质量守恒)



净流出: $[(\rho + \frac{\partial \rho}{\partial x} \frac{\Delta x}{2})(u + \frac{\partial u}{\partial x} \frac{\Delta x}{2}) \Delta y \Delta z - (\rho - \frac{\partial \rho}{\partial x} \frac{\Delta x}{2})(u - \frac{\partial u}{\partial x} \frac{\Delta x}{2}) \Delta y \Delta z]$
 $= \frac{\partial(\rho u)}{\partial x} \Delta x \Delta y \Delta z$
 同理在 y 向, z 向: $\frac{\partial(\rho v)}{\partial y} \Delta x \Delta z$ $\frac{\partial(\rho w)}{\partial z} \Delta x \Delta y$

单位时间由由于密度变化使控制体内的流体质量改变量: $\frac{\partial \rho}{\partial t} \Delta x \Delta y \Delta z$

质量守恒: $\frac{\partial \rho}{\partial t} \Delta x \Delta y \Delta z = - [\frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z}] \Delta x \Delta y \Delta z$

$\Rightarrow \frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} = 0$ $\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = 0$

三元定常: $\frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} = 0$

不可压缩 ($\rho = \text{const}$): $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$

柱坐标形式: $\frac{\partial \rho}{\partial t} + \frac{1}{r} [\frac{\partial(\rho r u)}{\partial r} + \frac{\partial(\rho v_\theta)}{\partial \theta} + \frac{\partial(\rho r w_z)}{\partial z}] = 0$

3. 定常流动中总流的连续性方程

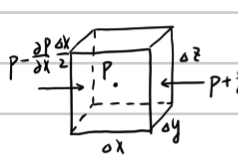
$-\int_{A_1} \rho u_n dA = \int_{A_2} \rho u_n dA$

流入流量 流出流量

平均速度 V_1, V_2 , 平均密度 $\rho_1, \rho_2 \Rightarrow \rho_1 V_1 A_1 = \rho_2 V_2 A_2$

$\rho = \text{const} \Rightarrow V_1 A_1 = V_2 A_2$

3.4 理想流体的运动微分方程 (动量方程)



总压力: $(p - \frac{\partial p}{\partial x} \frac{\Delta x}{2}) \Delta y \Delta z - (p + \frac{\partial p}{\partial x} \frac{\Delta x}{2}) \Delta y \Delta z = - \frac{\partial p}{\partial x} \Delta x \Delta y \Delta z$
 沿 x 方向质量力: $\rho f_x \Delta x \Delta y \Delta z$
 根据动量定理: $\rho \frac{du}{dt} \Delta x \Delta y \Delta z = \rho f_x \Delta x \Delta y \Delta z - \frac{\partial p}{\partial x} \Delta x \Delta y \Delta z$
 $\Rightarrow \frac{du}{dt} = f_x - \frac{1}{\rho} \frac{\partial p}{\partial x}$ 欧拉运动方程.

$$\begin{cases} \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} = f_x - \frac{1}{\rho} \frac{\partial p}{\partial x} \\ \frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} = f_y - \frac{1}{\rho} \frac{\partial p}{\partial y} \\ \frac{\partial w}{\partial t} + u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} = f_z - \frac{1}{\rho} \frac{\partial p}{\partial z} \end{cases} \Rightarrow \frac{\partial \vec{v}}{\partial t} + (\vec{v} \cdot \nabla) \vec{v} = \vec{f} - \frac{1}{\rho} \nabla p$$

3.5 理想流体定常运动伯努利方程

$$\begin{cases} u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} = -\frac{1}{\rho} \frac{\partial p}{\partial x} & ① \\ u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} = -\frac{1}{\rho} \frac{\partial p}{\partial y} & ② \\ u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} = -g - \frac{1}{\rho} \frac{\partial p}{\partial z} & ③ \\ \frac{dx}{u} = \frac{dy}{v} = \frac{dz}{w} \end{cases}$$

① · dx $\Rightarrow u \frac{\partial u}{\partial x} dx + u \frac{\partial v}{\partial y} dy + u \frac{\partial w}{\partial z} dz = -\frac{1}{\rho} \frac{\partial p}{\partial x} dx \Rightarrow u du = -\frac{1}{\rho} \frac{\partial p}{\partial x} dx$
 $v dv = -\frac{1}{\rho} \frac{\partial p}{\partial y} dy$
 $w dw = -g dz - \frac{1}{\rho} \frac{\partial p}{\partial z} dz$

三式相加 $\Rightarrow u du + v dv + w dw = -g dz - \frac{1}{\rho} (\frac{\partial p}{\partial x} dx + \frac{\partial p}{\partial y} dy + \frac{\partial p}{\partial z} dz)$

$g dz + \frac{1}{\rho} dp + d \frac{u^2 + v^2 + w^2}{2} = 0$ 对于不可压缩流体 ($\rho = \text{const}$) $\Rightarrow d(gz + \frac{p}{\rho} + \frac{u^2}{2}) = 0$ (u 表示沿流线方向的速度)

$$\Rightarrow \boxed{z + \frac{p}{\rho g} + \frac{u^2}{2g} = 0} \quad z_1 + \frac{p_1}{\rho g} + \frac{u_1^2}{2g} = z_2 + \frac{p_2}{\rho g} + \frac{u_2^2}{2g} \quad \text{伯努利方程.}$$

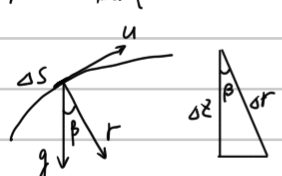
条件: ①定常 ②理想 ③ $\rho = \text{const}$ ④重力是唯一质量力

当流动无旋时, 方程在整个流场都成立.

2. 伯努利方程的意义

描述机械能守恒.

3.6 压强沿流线方向的变化



$$\begin{cases} a_r = f_r - \frac{1}{\rho} \frac{\partial p}{\partial r} & r \text{ 曲率半径.} \\ a_r = -\frac{u^2}{r} \end{cases} \Rightarrow -\frac{u^2}{r} = -\frac{d}{dr} \left(gz + \frac{p}{\rho} \right)$$

当 r 很大时, $-\frac{u^2}{r} \rightarrow 0$, 沿流向法向 r 有 $z + \frac{p}{\rho g} = C$

缓变流 — 流线曲率半径很大

急变流 — 流线曲率半径很小.

3.7 总流的伯努利方程.

假设 A_1, A_2 是缓变流

$$z_1 + \frac{p_1}{\rho g} + \frac{u_1^2}{2g} = z_2 + \frac{p_2}{\rho g} + \frac{u_2^2}{2g}$$

$$u_1 dA_1 = u_2 dA_2$$

$$\text{缓变流} \Rightarrow z_1 + \frac{p_1}{\rho g} = C, z_2 + \frac{p_2}{\rho g} = C$$

$$\int_{A_1} \left(z_1 + \frac{p_1}{\rho g} + \frac{u_1^2}{2g} \right) u_1 dA_1 = \int_{A_2} \left(z_2 + \frac{p_2}{\rho g} + \frac{u_2^2}{2g} \right) u_2 dA_2$$

$$\left(z_1 + \frac{p_1}{\rho g} \right) \int_{A_1} u_1 dA_1 + \int_{A_1} \frac{u_1^2}{2g} u_1 dA_1 = \left(z_2 + \frac{p_2}{\rho g} \right) \int_{A_2} u_2 dA_2 + \int_{A_2} \frac{u_2^2}{2g} u_2 dA_2$$

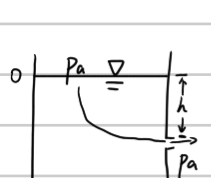
$$\text{截面平均速度 } V = \frac{1}{A} \int_A u dA$$

$$\int_A \frac{u^2}{2g} u dA = \alpha \frac{V^2}{2g} VA, \quad \alpha = \frac{1}{A} \int_A \left(\frac{u}{V} \right)^3 dA \quad \text{动能修正系数}$$

$$\text{总流的伯努利方程} \quad z_1 + \frac{p_1}{\rho g} + \alpha_1 \frac{V_1^2}{2g} = z_2 + \frac{p_2}{\rho g} + \alpha_2 \frac{V_2^2}{2g}$$

3.8 伯努利方程的应用

1. 小孔出流

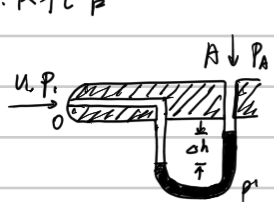


$$z_0 + \frac{p_0}{\rho g} = z_1 + \frac{p_1}{\rho g} + \alpha_1 \frac{V_1^2}{2g}$$

$$z_0 - z_1 = h, \quad \alpha_1 \approx 1$$

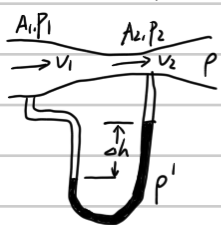
$$V_1 = \sqrt{2gh}$$

2. 皮托管



$$\begin{cases} \frac{p_0}{\rho} = \frac{p_A}{\rho} + \frac{u_A^2}{2} \\ p_A = p_1, u_A = V \end{cases} \Rightarrow V_1 = \sqrt{\frac{2(p_0 - p_1)}{\rho}} \quad p_0 - p_1 = (\rho' - \rho) g \Delta h \Rightarrow V_1 = \sqrt{\frac{2g(\rho' - \rho)}{\rho} \Delta h}$$

3. 文丘里流量计



$$\frac{P_1}{\rho} + \frac{v_1^2}{2} = \frac{P_2}{\rho} + \frac{v_2^2}{2}$$

$$v_2 = \sqrt{\frac{2(P_1 - P_2)}{\rho(1 - A_2^2/A_1^2)}}$$

3.9 叶轮机械内相对运动伯努力方程

$$z_1 + \frac{P_1}{\rho g} + \frac{u_1^2}{2g} - \frac{\omega^2 r_1^2}{2g} = z_2 + \frac{P_2}{\rho g} + \frac{u_2^2}{2g} - \frac{\omega^2 r_2^2}{2g}$$

$$H_1 = \left(z + \frac{P}{\rho g} + \frac{u^2}{2g} \right)_{r=R_1} \text{ 内轮处总水头}$$

$$H_2 = \left(z + \frac{P}{\rho g} + \frac{u^2}{2g} \right)_{r=R_2} \text{ 外轮处总水头}$$

$$H_2 = H_1 + \frac{\omega^2 (r_2^2 - r_1^2)}{2g}$$

不可压缩粘性流体的一元流动

4.1 粘性流体运动的伯努利方程

$$z_1 + \frac{p_1}{\rho g} + \frac{u_1^2}{2g} = z_2 + \frac{p_2}{\rho g} + \frac{u_2^2}{2g} + h'_{w1} \quad h'_{w1} \text{ — 从1到2所损失的机械能}$$

$$h_w = h_f + h_j \quad h_f = \lambda \frac{l}{d} \frac{V^2}{2g} \quad h_j = \zeta \frac{V^2}{2g}$$

4.2 层流与湍流

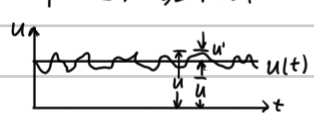
1. 流体运动的两种状态

平均速度 V 小于临界值 $V_c \rightarrow$ 层流, $h_f \propto V$
 大于 $V_c \rightarrow$ 湍流, $h_f \propto V^{1.75 \sim 2}$

V_c 非定值, $V_c \propto \frac{\mu}{\rho d}$

$$Re = \frac{\rho V d}{\mu} \quad \begin{cases} Re \leq 2300 & \text{层流} \\ Re > 2300 & \text{湍流} \end{cases}$$

2. 湍流运动的基本特征



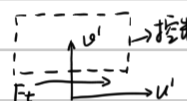
$$\bar{u} = \frac{1}{T} \int_{t-\frac{T}{2}}^{t+\frac{T}{2}} u dt \quad \bar{v} = \frac{1}{T} \int_{t-\frac{T}{2}}^{t+\frac{T}{2}} v dt \quad \bar{w} = \frac{1}{T} \int_{t-\frac{T}{2}}^{t+\frac{T}{2}} w dt \quad \bar{p} = \frac{1}{T} \int_{t-\frac{T}{2}}^{t+\frac{T}{2}} p dt$$

$$u = \bar{u} + u' \quad v = \bar{v} + v' \quad w = \bar{w} + w' \quad p = \bar{p} + p'$$

$$\bar{f}(x, y, z, t) = \frac{1}{T} \int_{t-\frac{T}{2}}^{t+\frac{T}{2}} f(x, y, z, t) dt \quad f(x, y, z, t) = \bar{f}(x, y, z, t) + f'(x, y, z, t)$$

时均值 脉动值 涨落

3. 湍流切应力及混合长理论

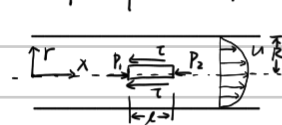


控制体A 动量关系: $F_t = -\rho Q(\bar{u} + u') = -\rho A \bar{u}'(\bar{u} + u')$
 除A取平均: $\frac{F_t}{A} = \tau_t = -\rho \bar{u}'\bar{u}' \quad \bar{u}\bar{u}' = \frac{1}{T} \int \left(\frac{u'}{T}\right) u dt = \frac{\bar{u}}{T} \int u' dt = 0$

层流中: $\tau = \mu \frac{du}{dy}$ 湍流中: $\tau = \mu \frac{du}{dy} - \rho \bar{u}'\bar{u}'$ 引入湍流粘度 μ_t $-\rho \bar{u}'\bar{u}' = \mu_t \frac{du}{dy}$

混合长理论: $-\rho \bar{u}'\bar{u}' = \rho l^2 \left| \frac{du}{dy} \right|^2 \quad -\rho \bar{u}'\bar{u}' = \rho l^2 \left| \frac{du}{dy} \right| \frac{du}{dy} \quad \mu_t = \rho l^2 \left| \frac{du}{dy} \right|$

4.3 圆管定常层流流动

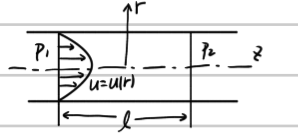


$$\tau = -\mu \frac{du}{dr} \quad (p_1 - p_2) \pi r^2 - 2\pi r l \tau = 0 \quad \left. \begin{matrix} r=R, u=0 \\ \Rightarrow u = -\frac{\Delta p}{4\mu} \frac{1}{l} (R^2 - r^2) \end{matrix} \right\} \Rightarrow u = -\frac{\Delta p}{4\mu} \frac{1}{l} (R^2 - r^2)$$

最大速度 $u_{max} = \frac{R^2 \Delta p}{4\mu l} \quad Q = \int_0^R u \cdot 2\pi r dr = \frac{\pi R^4 \Delta p}{8\mu l}$ 哈根-泊肃叶定理

平均速度 $V = \frac{Q}{A} = \frac{u_{max}}{2} \quad \Delta p = \frac{8\mu l V}{R^2} = \frac{32\mu l V}{d^2}$

动能修正系数 $z_1 + \frac{p_1}{\rho g} + \alpha_1 \frac{V_1^2}{2g} = z_2 + \frac{p_2}{\rho g} + \alpha_2 \frac{V_2^2}{2g} \quad \alpha = \frac{1}{A} \int_A \left(\frac{u}{V}\right)^2 dA = 2$



$$z_1 + \frac{p_1}{\rho g} + \alpha \frac{V^2}{2g} = z_2 + \frac{p_2}{\rho g} + \alpha \frac{V^2}{2g} + \lambda \frac{l}{d} \frac{V^2}{2g}$$

$$\frac{p_1 - p_2}{\rho g} = \frac{\Delta p}{\rho g} = \lambda \frac{l}{d} \frac{V^2}{2g} \quad \Delta p = \frac{32\mu l V}{d^2} \quad Re = \frac{\rho V d}{\mu}$$

$$\lambda = \frac{2d}{\rho V^2} \Delta p = \frac{64\mu}{\rho V d} = \frac{64}{Re}$$

1. 水力光滑管与水利粗糙管

粘性底层(近壁层流): $\frac{u_* y}{\nu} \leq 5 \quad u_* = \sqrt{\frac{\tau_w}{\rho}} \quad \mu \frac{du}{dy} \gg |-\rho \bar{u}'\bar{u}'|$

湍流核心区: $\frac{u_* y}{\nu} \geq 30 \quad \mu \frac{du}{dy} \ll |-\rho \bar{u}'\bar{u}'|$

过渡区: $5 < \frac{u_* y}{\nu} < 30 \quad \mu \frac{du}{dy} \sim |-\rho \bar{u}'\bar{u}'|$

水力光滑管: $\delta > 0$ 水力粗糙管: $\delta < 0$

粘性底层厚度(经验公式) $\delta = \frac{34.2d}{Re^{0.75}}$

2. 水利光滑管的速度分布和沿程损失系数

1) 粘性底层的速度分布

粘性底层切应力变化不大 $\tau_w = \mu \frac{du}{dy}$ 边界条件 $y=0$ 时 $u=0$ $u = \frac{\tau_w}{\mu} y$

定义摩擦速度 $u_* = \sqrt{\frac{\tau_w}{\rho}}$

$u = \frac{\tau_w}{\rho} \frac{y}{\nu} = u_*^2 \frac{y}{\nu}$

$\frac{u}{u_*} = \frac{u_* y}{\nu}$

2) 湍流核心区

$\mu \frac{du}{dy} \ll |\rho u' u'|$ 切应力变化不大 $\tau_w = -\rho u' u'$

混合长公式 $\tau_w = \rho l^2 \left(\frac{du}{dy} \right)^2$

$\frac{du}{dy} = \frac{1}{l} \sqrt{\frac{\tau_w}{\rho}} = \frac{u_*}{l}$ 设 $l = ky$, $\frac{du}{dy} = \frac{u_*}{ky} \rightarrow u = \frac{u_*}{k} \ln y + C$ 或 $\frac{u}{u_*} = \frac{1}{k} \ln \frac{u_* y}{\nu} + C$

边界条件: 粘性底层与湍流核心区交界处.

$\frac{u}{u_*} = \frac{1}{k} \ln \frac{u_* y}{\nu} + \frac{u_* \delta}{\nu} - \frac{1}{k} \ln \frac{u_* \delta}{\nu} = \frac{1}{k} \ln \frac{u_* y}{\nu} + B$

$\frac{u}{u_*} = 2.5 \ln \frac{u_* y}{\nu} + 5.5$

k - 卡门常数

(3) 水力光滑管沿程损失系数

$\frac{V}{u_*} = 2.5 \ln \left(Re \frac{u_*}{2V} \right) + 1.75$ 与 Δ 无关

$\frac{1}{\lambda} = 2 \lg (Re \sqrt{\lambda}) - 0.8$ 普朗特-施密特公式 $Re = 5 \times 10^4 \sim 3 \times 10^6$

$\lambda = \frac{0.3164}{Re^{0.25}}$ 布拉修斯公式 $Re = 3 \times 10^3 \sim 10^5$

(4) 水力粗糙管的沿程阻力系数

速度分布 $\frac{u}{u_*} = 2.5 \ln \frac{y}{\delta} + 8.5$ 与 Re 无关

平均速度 $\frac{V}{u_*} = 2.5 \ln \frac{R}{\delta} + 4.75$

损失系数 $\frac{1}{\lambda} = 2.05 \lg \frac{R}{\delta} + 1.697$

实验修正系数 $\frac{1}{\lambda} = 2 \lg \frac{d}{2\delta} + 1.74$

尼古拉兹图

1) 层流区 $Re < 2300$ λ 只与 Re 有关

2) 层流与湍流过渡区 $2300 < Re < 4000$ λ 只与 Re 有关, 与 Δ 无关

3) 水利光滑管区 $4000 < Re < 80 \frac{d}{\Delta}$ λ 只与 Re 有关, 与 Δ 无关 但此区上限取决于 $\frac{d}{\Delta}$, $h_f \propto V^{1.75}$

4) 水利光滑到粗糙过渡区 $80 \frac{d}{\Delta} < Re < 4160 \left(\frac{d}{2\delta} \right)^{0.85}$

5) 水利粗糙区 $Re > 4160 \left(\frac{d}{2\delta} \right)^{0.85}$

确定局部损失的方法

伯努力方程: $z_1 + \frac{P_1}{\rho g} + \frac{V_1^2}{2g} = z_2 + \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + h_j$

动量方程: $P_1 A_1 - P_2 A_2 = \rho Q (V_2 - V_1)$

1 连续性方程: $Q = V_1 A_1 = V_2 A_2$

$$h_j = \frac{(V_1 - V_2)^2}{2g} = \left(1 - \frac{A_1}{A_2}\right)^2 \frac{V_1^2}{2g} = \zeta_1 \frac{V_1^2}{2g} \quad \zeta_1 = \left(1 - \frac{A_1}{A_2}\right)^2$$
$$= \left(\frac{A_2}{A_1} - 1\right)^2 \frac{V_2^2}{2g} = \zeta_2 \frac{V_2^2}{2g} \quad \zeta_2 = \left(\frac{A_2}{A_1} - 1\right)^2$$

有压管水力计算

Chapter 5 可压缩流体的一元流动

5.1 基本公式

1. 状态方程

$$p = \rho R T$$

2. 连续性方程

$$\frac{dp}{\rho} + \frac{du}{u} + \frac{dA}{A} = 0$$

3. 运动方程

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} = f_x - \frac{1}{\rho} \frac{\partial p}{\partial x}$$

$$u \frac{du}{dx} = -\frac{1}{\rho} \frac{dp}{dx} \quad \text{或} \quad u du = -\frac{1}{\rho} dp$$

4. 热力学常数

$$\text{内能 } e = c_v T \quad \text{焓 } h = c_p T$$

$$c_v = \left(\frac{dq}{dT} \right)_v \quad c_p = \left(\frac{dq}{dT} \right)_p$$

定容热容 定压热容

$$\text{热容比 (绝热指数)} \gamma = \frac{c_p}{c_v}$$

$$\text{气体常数 } R = c_p - c_v = (\gamma - 1) c_v$$

$$c_v = \frac{R}{\gamma - 1} \quad c_p = \frac{\gamma R}{\gamma - 1}$$

5. 热力学第一定律 (能量守恒定律)

$$\delta q = de + p d\left(\frac{1}{\rho}\right)$$

$$e = c_v T \quad h = c_p T = (c_v + R)T = c_v T + RT = e + \frac{p}{\rho}$$

$$\delta q = de + d\left(\frac{p}{\rho}\right) - \frac{dp}{\rho} = dh + d\frac{u^2}{2}$$

对于绝热流动 $\delta q = 0$

$$\frac{u^2}{2} + h = C \quad \text{或} \quad \frac{u^2}{2} + c_p T = C \quad \frac{u^2}{2} + \frac{\gamma R}{\gamma - 1} T = C \quad \frac{u^2}{2} + \frac{\gamma}{\gamma - 1} \frac{p}{\rho} = C$$

6. 焓熵关系式

$$ds = \frac{\delta q}{T}$$

$$ds = \frac{de}{T} + \frac{p}{T} d\left(\frac{1}{\rho}\right) = c_v \frac{dT}{T} - R \frac{dp}{p}$$

$$s = c_v \int \frac{dT}{T} - R \int \frac{dp}{p} = c_v \ln T - R \ln p + C$$

$$= c_v \ln T - \gamma c_v \ln p + c_v \ln p + c_v \ln R - c_v \ln R + C$$

$$= c_v \ln \frac{p R T}{p^\gamma} - c_v \ln R + C$$

$$\frac{p}{\rho^\gamma} = C$$

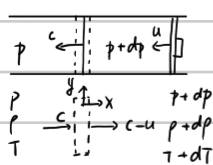
$$\frac{p_1}{p_2} = \left(\frac{\rho_1}{\rho_2} \right)^\gamma \quad \frac{p_1}{p_2} = \left(\frac{T_1}{T_2} \right)^{\frac{\gamma}{\gamma-1}} \quad \frac{p_1}{p_2} = \left(\frac{T_1}{T_2} \right)^{\frac{\gamma}{\gamma-1}}$$

5.2 微弱扰动波的传播及声速

1. 声波及声速

声波 — 微弱的压力(密度)扰动波

声速



$$\text{连续性方程 } \rho c A = (\rho + d\rho)(c - u) A \rightarrow u = \frac{d\rho}{\rho + d\rho} c$$

$$\text{动量方程 } p A - (p + dp) A = (\rho + d\rho)(c - u)^2 A - \rho c^2 A \rightarrow u = \frac{dp}{\rho c}$$

$$\frac{d\rho}{\rho + d\rho} c = \frac{dp}{\rho c} \rightarrow c^2 = \left(1 + \frac{d\rho}{\rho}\right) \frac{dp}{d\rho} \quad \text{略去小量 } \frac{d\rho}{\rho} \rightarrow c = \sqrt{\frac{dp}{d\rho}}$$

$$\text{在等熵条件下 } \frac{p}{\rho^\gamma} = c \rightarrow \frac{dp}{d\rho} = \gamma \frac{p}{\rho} \quad c = \sqrt{\gamma \frac{p}{\rho}} = \sqrt{\gamma R T}$$

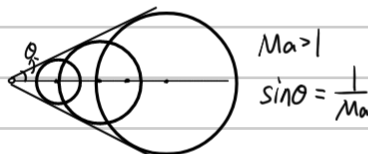
2. 马赫数

$$Ma = \frac{u}{c}$$

$$u du = -\frac{1}{\rho} d\rho$$

$$-u du = \frac{dp}{\rho} = \frac{dp}{\rho} \frac{dp}{d\rho} = \frac{dp}{\rho} c^2$$

$$-\frac{u^2}{c^2}$$



5.3 一元等熵流动基本关系

1. 滞止状态

气体流速为 0 的状态

$$\frac{u^2}{2} + c_p T = c \quad \text{动能为 0, 焓达最大值}$$

$$\frac{u^2}{2} + c_p T = c_p T_0 \quad c_p T_0 \text{ 为总能量, } T_0 \text{ 为总温}$$

$$\frac{T_0}{T} = 1 + \frac{u^2}{2} \frac{1}{c_p T} \quad \frac{T_0}{T} = 1 + \frac{\gamma - 1}{2} Ma^2$$

$$\frac{p_0}{p} = \left(1 + \frac{\gamma - 1}{2} Ma^2\right)^{\frac{1}{\gamma - 1}}$$

$$\frac{\rho_0}{\rho} = \left(1 + \frac{\gamma - 1}{2} Ma^2\right)^{\frac{\gamma}{\gamma - 1}}$$

2. 临界状态

$Ma = 1$ 临界参数 p^*, ρ^*, T^*, c^*

$$\frac{u^2}{2} + \frac{c^2}{\gamma - 1} = \frac{c_0^2}{2(\gamma - 1)} \quad c^2 = \frac{c_0^2}{\gamma + 1}$$

$$\frac{T^*}{T_0} = \frac{c^2}{c_0^2} = \frac{2}{\gamma + 1}$$

$$\frac{p^*}{p_0} = \left(\frac{2}{\gamma + 1}\right)^{\frac{1}{\gamma - 1}}$$

$$\frac{\rho^*}{\rho_0} = \left(\frac{2}{\gamma + 1}\right)^{\frac{\gamma}{\gamma - 1}}$$

3. 最大速度状态

$$\frac{u_{\max}^2}{2} = \frac{\gamma R}{\gamma - 1} T_0$$

$$c_p T + \frac{u^2}{2} = c_p T_0 = \frac{u_{\max}^2}{2}$$

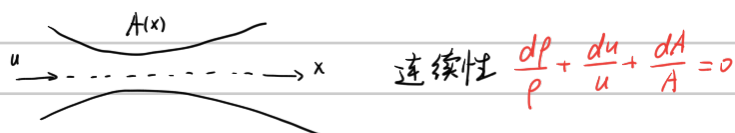
$$\frac{T}{T_0} = 1 - \left(\frac{u}{u_{\max}}\right)^2$$

$$\frac{p}{p_0} = \left[1 - \left(\frac{u}{u_{\max}}\right)^2\right]^{\frac{1}{\gamma - 1}}$$

$$\frac{\rho}{\rho_0} = \left[1 - \left(\frac{u}{u_{\max}}\right)^2\right]^{\frac{\gamma}{\gamma - 1}}$$

5.4 - 元步熵气流在变截面管中的流动

1. 气流参数与通道截面积之间的关系



运动方程 $u \frac{du}{dx} = -\frac{1}{\rho} \frac{dp}{dx} \quad u du = -\frac{1}{\rho} dp$

$$\frac{dp}{\rho} = \frac{dp}{dp} \frac{dp}{\rho} = \frac{1}{c^2} \frac{dp}{\rho} = -\frac{u}{c^2} du = -Ma^2 \frac{du}{u}$$

$$(Ma^2 - 1) \frac{du}{u} = \frac{dA}{A}$$

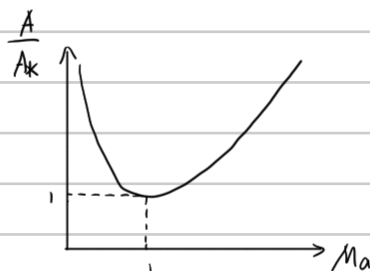
① 亚声速 A 增 u 减, A 减 u 增.

② 超声速 A 增 u 增, A 减 u 减.

③ $Ma=1$ $dA=0$ 声速只存在截面极小值处.

$$\frac{1-Ma^2}{Ma^2} \frac{dp}{p} = \frac{dA}{A} \quad \frac{1-Ma^2}{\gamma Ma^2} \frac{dp}{p} = \frac{dA}{A} \quad \frac{1-Ma^2}{(\gamma-1)Ma^2} \frac{dT}{T} = \frac{dA}{A}$$

$$\frac{A_2}{A_1} = \frac{Ma_1}{Ma_2} \left[\frac{2+(\gamma-1)Ma_2^2}{2+(\gamma-1)Ma_1^2} \right]^{\frac{\gamma+1}{2(\gamma-1)}} \quad \frac{A}{A_*} = \frac{1}{Ma} \left[\frac{2+(\gamma-1)Ma^2}{\gamma+1} \right]^{\frac{\gamma+1}{2(\gamma-1)}}$$



2. 收缩喷管.

$$c_p T_c + \frac{u_c^2}{2} = c_p T_0$$



$$u_c = \sqrt{2c_p(T_0 - T_c)} = \sqrt{2c_p T_0 \left(1 - \frac{T_c}{T_0}\right)} \quad \frac{T_c}{T_0} = \left(\frac{p_c}{p_0}\right)^{\frac{\gamma-1}{\gamma}}$$

$$u_c = \sqrt{2c_p T_0 \left[1 - \left(\frac{p_c}{p_0}\right)^{\frac{\gamma-1}{\gamma}}\right]} \quad \frac{p}{p_0} = \left(\frac{p}{p_0}\right)^{\frac{\gamma-1}{\gamma}}$$

$$Q_m = p_0 A_e \left(\frac{p_c}{p_0}\right)^{\frac{1}{\gamma}} \sqrt{2c_p T_0 \left[1 - \left(\frac{p_c}{p_0}\right)^{\frac{\gamma-1}{\gamma}}\right]}$$

背压 p_b 当 $p_b > p_*$, $p_e = p_b$

当 $p_b = p_*$, $p_e = p_*$

3. 缩放喷管